

Jordan Chen, Ph.D.

✉ jordan.chen@email.com

📍 Richland, WA | [in LinkedIn](#) | [ORCID: 0000-0002-1234-5678](#) | [Google Scholar](#)

Research Summary

Computational biologist with 8+ years spanning molecular dynamics simulation and ML-guided protein engineering, bridging biomolecular simulation and data-driven design for enzyme engineering and drug discovery. Fine-tuned the ESM-2 protein language model on 45K experimental stability measurements to screen 8,500 enzyme variants at 3,000× experimental throughput, with 5 hits confirmed by collaborators via differential scanning calorimetry. Co-developed an open-source transfer learning framework adopted by 4 external research groups, and built enhanced-sampling protocols for protein folding and ligand binding that now underpin the group's simulation infrastructure. Author of 15 publications (7 first-author, 280+ citations) in venues including ACS Catalysis, J. Chem. Theory Comput., and J. Med. Chem.

Education

Ph.D., Biomedical Engineering Aug 2018 – Jul 2023
Westfield Institute of Technology, Westfield, MA GPA: 3.92/4.00
Dissertation: Enhanced Sampling Methods for Protein Folding and Ligand Binding Free Energy Prediction
Advisor: Prof. L. Alvarez

B.S., Biochemistry (Honors) Aug 2014 – May 2018
Eastgate University, Portland, OR GPA: 3.87/4.00

Technical Expertise

Molecular Simulation & Modeling

- GROMACS, OpenMM, AMBER – metadynamics, replica exchange MD, free energy perturbation
- AlphaFold2, Rosetta, AutoDock Vina – structure prediction, homology modeling, docking
- CHARMM36m, AMBER ff19SB, OPLS-AA/M – force field benchmarking for disordered proteins

Machine Learning & Data Science

- Protein language models (ESM-2, 650M), graph neural networks, transfer learning
- PyTorch, scikit-learn, BioPython – fine-tuning, embedding extraction, feature engineering
- Regression, classification, cross-validation, Spearman/RMSE benchmarking, dataset curation

Analysis & Visualization

- MDAnalysis, ProDy, PyMOL, matplotlib, seaborn – trajectory analysis, structural visualization
- PostgreSQL, pandas, NumPy – curated stability databases with automated quality filters

Domain Expertise

- Protein engineering, enzyme thermostability, folding thermodynamics, drug discovery
- Intrinsically disordered proteins, ligand binding free energy, biocatalysis, directed evolution

Programming & HPC

- Python, Bash, SQL – scientific computing, data pipelines, automated analysis workflows
- SLURM, Snakemake, Git, DVC – HPC job scheduling, workflow automation, reproducible research

Research Experience

Postdoctoral Research Associate Aug 2023 – Present
Lakewood University, Department of Bioengineering
PI: Prof. K. Holmberg

ML-Guided Enzyme Stability Screening

- Fine-tuned the ESM-2 protein language model on 45K experimental melting temperatures, achieving 0.82 Spearman correlation and enabling $3,000\times$ throughput screening of 8,500 enzyme variants.
- Co-developed a transfer learning framework reducing labeled training data requirements by 60% across five enzyme families, released as an open-source tool with 200+ GitHub stars.

Enzyme Solvent Tolerance for Green Chemistry

- Extended the protein language model to predict enzyme solvent tolerance across eight organic co-solvent systems, validating against 50-ns explicit-solvent MD for 80 enzyme variants.

Automated Simulation Infrastructure

- Automated the sequence-to-simulation pipeline using the Snakemake workflow manager, reducing per-variant setup time from 4 hours to 10 minutes for six researchers.

Ph.D. Researcher

Aug 2018 – Jul 2023

Westfield Institute of Technology

Advisor: Prof. L. Alvarez

Enhanced Sampling for Protein Folding

- Developed a metadynamics-based enhanced sampling protocol for protein folding free energy landscapes, predicting folding temperatures within 8 K of experiment across six proteins.

Ligand Binding Free Energy Calculations

- Calculated relative binding free energies for 40 congeneric ligand pairs via free energy perturbation, achieving 0.9 kcal/mol RMSE against experimental IC₅₀ data.

Thermostability Database Infrastructure

- Built a curated protein thermostability database integrating 12,000 experimental melting temperatures from 3 public sources, with quality filters adopted by 8 lab members.

Undergraduate Research Assistant

May 2016 – Jul 2018

Eastgate University

- Performed homology modeling and 100-ns MD simulations of 4 mutant lysozyme variants, identifying destabilizing cavity mutations consistent with published experimental data.
- Built Python analysis scripts for automated hydrogen bond occupancy tracking across 500-ns aggregate trajectories, adopted by 3 lab members for stability projects.

Fellowships & Honors

- **NSF Graduate Research Fellowship**, National Science Foundation (2019). Three-year fellowship supporting doctoral research in computational protein engineering.
- **Best Oral Presentation**, Westfield Biophysics Symposium (2022). Enhanced sampling methods for protein folding thermodynamics.
- **Dean's Teaching Award**, Westfield Institute of Technology (2021). Outstanding TA in computational biology.
- **Outstanding Poster Award**, Gordon Research Conference on Proteins (2023). ML-guided enzyme thermostability screening.

Publications

Selected publications shown below; full list available on Google Scholar.

Published

6. **J. Chen**, R. Nakamura, S. Patel, K. Holmberg, M. Rivera. “Deep Learning-Guided Screening of Thermostable Enzyme Variants for Industrial Biocatalysis.” *ACS Catalysis* (2025).
5. **J. Chen**, M. Rivera, K. Holmberg. “Transfer Learning from Protein Language Models for Low-Data Enzyme Property Prediction.” *Bioinformatics* (2024).
4. **J. Chen**, L. Alvarez. “Ligand Binding Free Energies via Enhanced-Sampling FEP for Three Drug Target Families.” *J. Med. Chem.* (2023).
3. **J. Chen**, L. Alvarez. “Metadynamics Protocol for Protein Folding Free Energy Landscapes.” *J. Chem. Theory Comput.* (2022).
2. **J. Chen**, P. Kowalski, L. Alvarez. “Force Field Benchmarking for Intrinsically Disordered Protein Ensembles.” *J. Chem. Theory Comput.* (2021).
1. **J. Chen**, P. Kowalski, L. Alvarez. “Curated Thermostability Database for ML-Ready Protein Engineering Benchmarks.” *Bioinformatics* (2021).

Under Review

- **J. Chen**, T. Yamamoto, K. Holmberg. “Predicting Enzyme Solvent Tolerance with Fine-Tuned Protein Language Models.” *Proteins: Struct., Funct., Bioinf.* Under review.

Selected Presentations

- **J. Chen**. “ML-Guided Screening of Thermostable Enzyme Variants.” *Gordon Research Conference on Proteins*, Ventura, CA (2023). Poster.
- **J. Chen**, M. Rivera, K. Holmberg. “Transfer Learning from Protein Language Models for Low-Data Enzyme Property Prediction.” *ACS National Meeting*, San Francisco, CA (2024). Oral.
- **J. Chen**, L. Alvarez. “Enhanced Sampling Protocols for Protein Folding Free Energy Landscapes.” *Biophysical Society Annual Meeting*, San Diego, CA (2022). Oral.
- **J. Chen**, P. Kowalski, L. Alvarez. “Force Field Benchmarking for Intrinsically Disordered Protein Ensembles.” *AIChE Annual Meeting*, Boston, MA (2021). Poster.
- **J. Chen**. “Automated Screening Pipelines for ML-Guided Enzyme Engineering.” *Lakewood University Computational Biology Seminar Series* (2024). Invited talk.

Mentorship & Professional Service

- **Graduate mentorship:** Mentored 3 graduate students on protein ML pipelines and MD simulation workflows; one student co-authored a peer-reviewed publication within 8 months of joining.
- **Teaching:** Recognized with the Dean’s Teaching Award (2021) for outstanding TA performance in computational biology coursework.
- **Lab training adoption:** Analysis scripts and quality-filter protocols adopted by 3–8 lab members across three research groups for ongoing protein stability projects.